EG2605 UNDERGRADUATE RESEARCH OPPORTUNITIES PROGRAMME (UROP)

UROP REPORT

Modelling of Emerging Contaminants in Singapore's Coastal Waters: Exploring the Relationship between the Taxonomic Composition of Contaminants and their Antibiotic Resistance Genes (ARGs) by using Machine Learning

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Abstract

This research explores the relationship between the taxonomy composition and the antibiotic resistance genes (ARGs) of Enterococcus sampled from Singapore's coastal water using machine learning. The research aims to identify potential antimicrobial resistance (AMR) in Enterococcus. Most importantly, the research aims to enhance the efficiency of identifying AMR in water sources while maintaining accuracy.

The research motivation is reinforced through a literature review, which highlights the significance of water quality, emerging contaminants, and the role of bioinformatics and machine learning in AMR detection. Subsequently, the methodology is demonstrated, including cultivation, DNA extraction, sequencing, and modelling. Notably, the Naïve Bayes model is employed to explore associations between Enterococcus taxonomy and ARGs, particularly those related to Vancomycin resistance. Results indicate a strong association between taxonomic composition and ARGs and a model accuracy of 96.15%. Following that, limitations of the model are addressed, including limited generalizability, the risk of overfitting, and bias towards specific resistance mechanisms.

Overall, the research provides valuable insights into AMR dynamics and underscores the importance of advanced computational methods in mitigating antibiotic resistance.

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Together, Professor Gin and my mentors have created an environment conducive to learning and growth, for which I am truly thankful. Their mentorship has not only enhanced my research skills but has also inspired me to strive for excellence in all my academic pursuits. I am privileged to have had the opportunity to work alongside such dedicated and supportive individuals.

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1. Introduction

Water is undeniably one of humanity's most important sources of nutrients. It plays a pivotal role in transporting nutrients to cells, eliminating waste, and regulating body temperature (UC Davis Health, 2022). Evidently, a continual supply of clean water is essential for maintaining human health and enhancing quality of life. Thus, the quality of the water supply is very important.

However, there are emerging contaminants in water bodies that have antimicrobial resistance (AMR) (Usui et al., 2022). These contaminants, known as antibiotic-resistant bacteria (ARB), carry antibiotic resistance genes (ARG), enabling them to withstand antibiotic medications. Hence, consumption of water contaminated with such bacteria can lead to illnesses unresponsive to antibiotics. In fact, it is projected that AMR could become the leading cause of death by 2050 (Stanton et al., 2021). Therefore, the issue of ARB in water sources is crucial due to its potential impact on human health.

Nowadays, identifying ARG in bacteria, meaning ARB, uses bioinformatics, specifically the Basic Logic Alignment Search Tool (BLAST). However, it requires many datasets as input and also consumes a large amount of time (Wikipedia, n.d.). Such requirements may impede the efficiency of obtaining insights into water quality, consequently delaying preventive and mitigation measures for inadequate water quality.

Hence, in this research, **I will be employing machine learning methods to model the relationship between the taxonomic composition of ARB and its associated ARG.** This approach aims to identify any possible associations in their relationship. Subsequently, by integrating machine learning into bioinformatics, the efficiency of water quality testing could perhaps be enhanced while still maintaining accuracy.

In this report, I will be focusing on Enterococcus bacteria and its associated ARG. The Enterococcus samples were obtained from Singapore's coastal water, specifically an open-caged sea farm at Lim Chu Kang. The samples were provided by the Singapore Food Agency (SFA), an agency that sets stringent food standards for Singapore

(Singapore Food Agency, n.d.). Basically, it ensures the health of people by monitoring the cleanliness of food and water. In this research, samples were provided to evaluate the quality of the water.

I will first start by doing a literature review to further reinforce my research motivation. Subsequently, I will demonstrate the methodology used, including the laboratory preparation and contaminant modelling. After that, I will provide visualizations of the model built, followed by an evaluation of the model and the methodology. Lastly, the report will end with some closing remarks.

2. Literature Review

In this section, I will delve into a more in-depth literature review to reinforce my motivation for **exploring the relationship between the taxonomy composition and the antibiotic resistance genes of Enterococcus sampled from Singapore's coastal water using machine learning.** Firstly, I plan to cover Singapore coastal water as the macroenvironmental background of my research, Subsequently, I will delve into the target bacteria, Enterococcus, and inspect its taxonomy. Following that, I will research the characteristics of antimicrobial resistance (AMR), antibiotic-resistant bacteria (ARB), and antibiotic resistance genes (ARG), as well as their causes of emergence. Lastly, I will briefly touch on BLAST in bioinformatics, and decide on the machine learning model that I will be utilizing in this research.

2.1 Singapore and Coastal Water

Singapore is located between the South China Sea and the Malacca Straits (Schmoker et al., 2014). This geographical positioning has placed Singapore in a very advantageous position for trading, and hence Singapore has been able to develop at an accelerated pace to reach its current level of development. However, due to its land size of 720 square kilometers, Singapore has limited space for agriculture (Singapore Academy, n.d.). Hence, Singapore has always faced the problem of food security. For example, the COVID-19 pandemic has exposed vulnerabilities in Singapore's food supply chain (Teng, 2020). As the country heavily relies on imported foods, Singapore's development was constrained during the pandemic period due to the food supply. Thus, the government initiated the 30 by 30 goal, specifically, to increase local food production capacity so that it reaches 30% of total nutritional needs by 2030 (Singapore Food Agency, n.d.). So far, many initiatives have been undertaken, including vertical farming for vegetables and offshore fish farming. In this research, offshore fish farming is the subject of interest, as the samples of Enterococcus came from an open-caged fish farm in Lim Chu Kang. In 2022, these fish farms produced a total of 4400 tons of seafood, accounting for a significantly low 8 percent of all seafood consumed (Begum, 2023). However, even though the number is relatively low, there is still a need to maintain good

water quality. This is to ensure that the supply of seafood can readily increase so that Singapore's food security goals can be achieved. Hence, it is important to monitor the condition of bacteria like Enterococcus in the coastal water.

2.2 Enterococcus and Taxonomy

Enterococci are a group of Gram-positive bacteria that are ubiquitous in nature and are commonly found in various environments, including soil, water, plants, and the gastrointestinal tracts of humans and animals (García-Solache & Rice, 2019). The taxonomy of Enterococci has undergone several revisions over the years (Kalina, 1970). Enterococci were initially described as *Streptococcus faecalis* (Andrewes & Horder, 1906). Enterococci were later reclassified into the genus Enterococcus. The type species of the genus Enterococcus was proposed to be *Enterococcus proteiformis*, but subsequent research and investigations led to the proposal of *Enterococcus faecalis* as the type species, as shown in **Figure 1**. Further taxonomic studies have also identified additional species within the genus Enterococcus, including *Enterococcus faecium* (Orla-Jensen, 1919). This species has been extensively studied and found to exhibit various subspecies, such as *Enterococcus faecium subsp. durans.* Enterococcus have gained significant attention in recent years due to their clinical relevance as opportunistic pathogens. Further research is needed to better understand the taxonomy, epidemiology, and pathogenicity of Enterococcus to develop effective strategies for their control and management in both clinical and environmental settings.

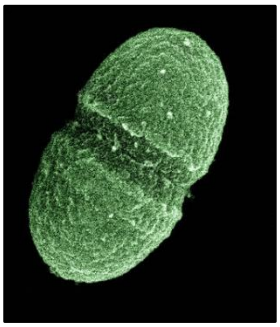

Figure 1: *Enterococcus faecalis* (Wikipedia, n.d.)

2.3 AMR, ARG, and ARB

Antimicrobial resistance (AMR) is a global health challenge. It enables bacteria to adapt and survive exposure to antimicrobial agents. Across various bacterial species, including Enterococcus, AMR emerges through the acquisition or mutation of genes encoding resistance mechanisms, known as antibiotic resistance genes (ARGs). These genes are the root cause of the ability of bacteria, or antibiotic-resistant bacteria (ARB), to withstand the effects of antimicrobial drugs. Enterococcus species, notably *Enterococcus faecalis* and *Enterococcus faecium*, are of particular concern. According to Zaheer et al. (2020) and Ikhimiukor et al. (2022), this is due to Enterococcus intrinsic resistance to several antimicrobial classes and their ability to acquire additional resistance genes through horizontal gene transfer.

Furthermore, Ikhimiukor et al. delve into the transmission dynamics of AMR in low- and middle-income countries (LMICs) settings. They emphasized the challenges posed by rapid urbanization and informal urban environments. They also highlighted the role of dense transmission networks, poor water and sanitation infrastructure, and resource extraction practices, all of which are amplifying transmission risks among human, animal, and environmental populations. In these settings, ARBs and ARGs can circulate undetected. Although Singapore falls outside the scope of LMICs, its journey to development highlights the importance of continued research. Fortunately, much water quality monitoring research is being initiated to ensure that there is adequate water and sanitation infrastructure.

2.4 BLAST and Machine Learning

The Basic Local Alignment Search Tool (BLAST) is an important technology in bioinformatics that is usually executed after the sequencing of genetic material. BLAST operates by comparing query sequences to databases or libraries. Then, it uses heuristic algorithms to identify regions of local similarity, known as high-scoring segment pairs (HSPs) and maximal-scoring segment pairs (MSPs) (Pertsemlidis & Fondon, 2001). In this process, it identifies closely related genes (Feldgarden et al., 2019). However,

despite its widespread use and effectiveness, BLAST encounters computational challenges. For example, the Smith-Waterman algorithm, known for its accuracy in sequence alignment, exhibits a quadratic time complexity of O(n^2), where 'n' represents the length of the sequences under comparison (Brown University Department of Computer Science, 2022). This computational challenge becomes evident, especially when handling large datasets. BLAST also requires substantial memory resources for storing and processing sequences. Thus, while BLAST remains a fundamental tool in bioinformatics for its accuracy, its computational demands underscore the ongoing need for optimization strategies to enhance efficiency and scalability.

Machine learning presents a promising approach to alleviate the computational burdens of BLAST and enhance its efficiency in bioinformatics tasks. Specifically, it offers avenues to model complex relationships using classification modeling techniques. Machine learning-based classification models such as random forest, k-nearest neighbor (KNN), and logistic regression could offer more streamlined and scalable approaches to analyzing the genes. For example, a random forest model could capture intricate taxonomy-gene patterns by aggregating decision trees (Garg, 2018). A KNN model could identify gene associations by considering taxonomic similarities among samples. A logistic regression model could provide interpretable insights into taxonomy-gene relationships. Overall, machine learning-based classification modelling might be able to increase efficiency in identifying closely related genes.

3. Methodology

3.1 Cultivation

The cultivation of Enterococcus is an initial step aimed at preparing the bacteria for further experimental processes. Enterococcus species are commonly found in various environments and are known for their ability to survive in diverse conditions. Cultivating Enterococcus allows researchers to study its growth patterns, metabolic activities, and unlock insights into its genetic makeup. This method delineates the necessary steps to initiate Enterococcus growth in a controlled laboratory setting, facilitating subsequent DNA extraction procedures. In this research, the cultivation of Enterococcus is executed as follows:

1. Using a sterile pipette, dispense 2000µL of Brain Heart Infused Broth (BHIB) into a well of a sterile culture plate. BHIB serves as a nutrient-rich medium conducive to Enterococcus growth.
2. From a centrifuge tube containing Enterococcus cultures, pipette 10µL of the bacterial suspension into the well, as shown in **Figure 2**.
3. Repeat step 1 & 2 for 27 samples to ensure consistency and reliability across experiments.
4. Clearly label each well of the culture plate to correspond with the respective cultures, ensuring accurate identification.
5. Seal the culture plate securely with parafilm to safeguard against contamination during incubation.
6. Place the sealed culture plate in an incubator set to 37°C for 8 hours (overnight) to facilitate optimal growth conditions for Enterococcus, as shown in **Figure 3**.
7. Following the incubation period, visually assess the wells for signs of turbidity, indicative of Enterococcus growth.

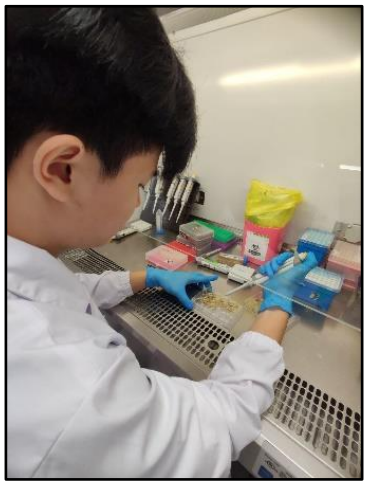

Figure 2: Pipetting Enterococcus

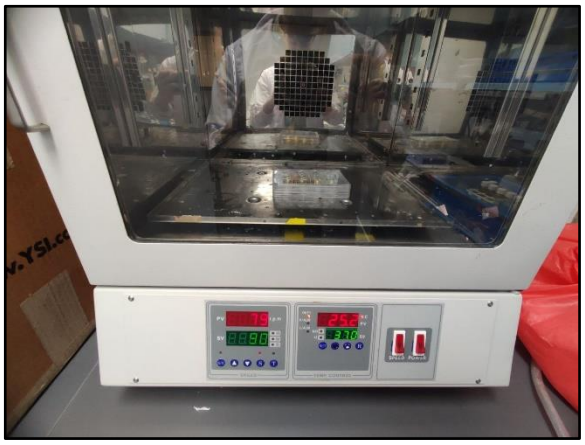

Figure 3: Incubation

3.2Extraction

The DNA extraction of Enterococcus is a crucial procedure following cultivation, which enables researchers to delve deeper into the genetic composition of this bacterium. Understanding their genetic composition is essential for applications in various fields, such as the study of their physiological traits and pathogenicity. This method facilitates downstream analysis, including sequencing and genetic engineering. In this research, the DNA extraction of Enterococcus is executed by following the instructions in the DNeasy PowerSoil Pro Kit Handbook.

Following that, the quality of the extracted DNA is assessed using a nanodrop spectrophotometer. As shown in **Figure 4**, most samples (excluding blank) have a concentration above 30 ng/µL. Additionally, most samples have an absorbance 260/280 ratio of around 1.8, while some samples have an absorbance 260/230 ratio greater than 1.5; hence, the samples are at an adequate level for sequencing (Marg-Haufe, n.d.).

#	Sample ID	User name	Date and Time	Nucleic Acid	Unit	A260 (Abs)	A280 (Abs)	260/280	260/230	Sample Type	Factor
1	E1	9020user	2024-03-07 14:26:45	37.1	ng/µl	0.742	0.384	1.93	1.77	DNA	50.00
2	E2	9020user	2024-03-07 14:28:48	35.6	ng/µl	0.711	0.371	1.92	0.81	DNA	50.00
3	E3	9020user	2024-03-07 14:30:22	53.3	ng/µl	1.066	0.555	1.92	2.02	DNA	50.00
4	E4	9020user	2024-03-07 14:31:36	44.9	ng/µl	0.898	0.469	1.91	1.63	DNA	50.00
5	E5	9020user	2024-03-07 14:32:43	60.0	ng/µl	1.199	0.629	1.91	1.36	DNA	50.00
6	E6	9020user	2024-03-07 14:33:59	42.7	ng/µl	0.853	0.440	1.94	0.51	DNA	50.00
7	E7	9020user	2024-03-07 14:34:56	55.0	ng/µl	1.100	0.574	1.92	1.04	DNA	50.00
8	E8	9020user	2024-03-07 14:36:01	47.8	ng/µl	0.955	0.499	1.91	0.82	DNA	50.00
9	E9	9020user	2024-03-07 14:37:22	49.8	ng/µl	0.996	0.517	1.93	1.91	DNA	50.00
10	E10	9020user	2024-03-07 14:39:16	62.3	ng/µl	1.247	0.657	1.90	1.89	DNA	50.00
11	E11	9020user	2024-03-07 14:40:19	55.8	ng/µl	1.117	0.581	1.92	2.04	DNA	50.00
12	E12	9020user	2024-03-07 14:41:24	43.6	ng/µl	0.873	0.463	1.89	1.63	DNA	50.00
13	E13	9020user	2024-03-07 14:42:44	48.3	ng/µl	0.966	0.501	1.93	1.82	DNA	50.00
14	E14	9020user	2024-03-07 14:43:47	22.0	ng/µl	0.441	0.216	2.04	0.07	DNA	50.00
15	E15	9020user	2024-03-07 14:45:06	34.6	ng/µl	0.692	0.353	1.96	0.16	DNA	50.00
16	E16	9020user	2024-03-07 14:46:06	20.6	ng/µl	0.411	0.206	2.00	0.70	DNA	50.00
17	E17	9020user	2024-03-07 14:47:03	24.0	ng/µl	0.481	0.247	1.95	1.38	DNA	50.00
18	E18	9020user	2024-03-07 14:47:52	65.8	ng/µl	1.315	0.687	1.92	0.23	DNA	50.00
19	E19	9020user	2024-03-07 14:48:49	20.4	ng/µl	0.408	0.201	2.03	0.05	DNA	50.00
20	E20	9020user	2024-03-07 14:49:40	21.9	ng/µl	0.438	0.229	1.91	0.30	DNA	50.00
21	E21	9020user	2024-03-07 14:50:31	18.5	ng/µl	0.369	0.187	1.97	0.45	DNA	50.00
22	E22	9020user	2024-03-07 14:52:54	19.8	ng/µl	0.396	0.204	1.94	0.25	DNA	50.00
23	E23	9020user	2024-03-07 14:53:38	32.3	ng/µl	0.645	0.343	1.88	1.18	DNA	50.00
24	E24	9020user	2024-03-07 14:54:19	11.3	ng/µl	0.227	0.113	2.01	0.34	DNA	50.00
25	E25	9020user	2024-03-07 14:55:08	17.3	ng/µl	0.345	0.186	1.86	0.38	DNA	50.00
26	E26	9020user	2024-03-07 14:55:57	168.3	ng/µl	3.367	1.784	1.89	1.74	DNA	50.00
27	E27	9020user	2024-03-07 14:56:50	14.5	ng/µl	0.291	0.152	1.92	0.67	DNA	50.00
28	B	9020user	2024-03-07 14:57:43	0.8	ng/µl	0.017	0.014	1.24	0.05	DNA	50.00

Figure 4: DNA Concentration

3.3 Sequencing

The sequencing of Enterococcus DNA is executed immediately after DNA extraction to provide insights into the genetic makeup of Enterococcus. This process is also to gather information for subsequent analyses and interpretations; in this research, it is also for modelling. The sequencing procedure in the research is outsourced to Novogene.

Following the sequencing phase, a series of pre-analysis processing steps are undertaken to prepare the raw sequencing data for downstream analysis. These steps are executed under the guidance of Eric. It includes trimming to remove low-quality base pairs. Quality assessment and error checking to ensure data integrity. Subsequently, assembly is needed to reconstruct the DNA sequence. Lastly, taxonomic classification is used to identify Enterococcus species. After this step, BLAST analysis is used to detect ARGs within the sequenced fragments. In this research, BLAST is not only used to **detect AMR behaviors in Enterococcus**. It also acts as a calibrating standard to **evaluate the accuracy of machine learning models**.

3.4 Modelling

The modelling of Enterococcus in this research is to **explore potential relationships between its taxonomic composition and ARG**. Following pre-analysis processing steps, sample E27 and sample B (blank) are excluded due to their taxonomic classification not being Enterococcus, leaving 26 remaining datasets. After that, R Studio is used to further filter and select important variables. This method involves the TidyVerse package and the StringR package, which facilitate data presentation in a tidy format, as shown in **Figure 5**. In this dataset, "Name", "Taxonomy", and "ARGs" are the main variables, as shown in **Figure 6** (note that there are in total 46 different ARGs identified using BLAST). Subsequently, the data is prepared for modelling.

```
12 ```{r warning = FALSE}
13 library(tidyverse)
14 library(stringr)
15 ```
16
17 # ARG
18 ```{r warning = FALSE}
19 data_list = list()
20 file_paths = paste0("C:/Users/yihah/OneDrive/Documents/NUS/MISC/UROP/Data/E", 1:26, ".tsv")
21
22 for (i in seq_along(file_paths)) {
23   data_list[[i]] = read.delim(file_paths[i])
24 }
25
26 names(data_list) = paste0("E", 1:26)
27 ```
28
29 # Taxonomic Composition
30 ```{r warning = FALSE}
31 Taxonomic_Composition = read.delim("C:/Users/yihah/OneDrive/Documents/NUS/MISC/UROP/Data/ent.bac120.summary.tsv")
32
33 Taxonomic_Composition = Taxonomic_Composition %>%
34   select(user_genome, classification) %>%
35   filter(!(user_genome %in% c("B", "E27")))
36 ```
37
38 # Combining Tables
39 ```{r warning = FALSE}
40 Combined_Table = do.call(rbind, data_list)
41
42
43 Combined_Table = Combined_Table %>%
44   select(Name, ARG = Gene.symbol) %>%
45   left_join(Taxonomic_Composition,c("Name" = "user_genome")) %>%
46   arrange(as.numeric(sub("^E", "", Name))) %>%
47   rename(Taxonomy = classification) %>%
48   mutate(Taxonomy = str_extract(Taxonomy, "(?<=;)s__.+$")) %>%
49   pivot_wider(names_from = ARG, values_from = ARG, values_fn = length, values_fill = 0)
50 ```
```

Figure 5: Code Chunk for Data Presentation

	Name	Taxonomy	lsa	copB	copA	tet(M)	aac(6')-I	abc-f	fosX	cadC	cadA	vanC	vanXY-C	vanTc	vanR-C	vanS-C	arsB	sal(A)	fosB
1	E1	s__Enterococcus_B hirae	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
2	E2	s__Enterococcus_B faecium	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
3	E3	s__Enterococcus_D casseliflavus	0	0	0	1	0	2	1	1	1	1	1	1	1	1	0	0	0
4	E4	s__Enterococcus_B hirae	1	1	1	1	1	3	1	1	1	1	1	1	1	1	2	1	1
5	E5	s__Enterococcus_B hirae	1	1	1	0	1	1	0	0	0	0	0	0	0	0	2	1	1
6	E6	s__Enterococcus faecalis	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0
7	E7	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
8	E8	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
9	E9	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
10	E10	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
11	E11	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
12	E12	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
13	E13	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
14	E14	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
15	E15	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
16	E16	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
17	E17	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
18	E18	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
19	E19	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
20	E20	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
21	E21	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
22	E22	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
23	E23	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
24	E24	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
25	E25	s__Enterococcus faecalis	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
26	E26	s__Enterococcus_B lactis	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0

Figure 6: Data Processed for Modelling (Partial)

The first step in modelling is to establish the association between Enterococcus's taxonomy and ARG. However, as there are too many different ARGs, it could be difficult to express a clear association. Therefore, to streamline the analysis, the significant ARGs are kept while the insignificant ARGs are removed.

In this research, the significant and more interesting ARGs would be the Vancomycin-resistant genes, as the Vancomycin antibiotic is usually effective against Gram-positive bacteria, like Enterococcus (Reynolds, n.d.). Vancomycin resistance signifies a crucial aspect of bacterial resistance, especially in the context of clinical treatment. Consequently, ARGs associated with Vancomycin resistance are more worthy of further investigation and study. Thus, a one-hot encoding approach is employed to transform categorical variables into a binary format, where ARGs related to Vancomycin resistance are represented as 1, while others are denoted as 0, as shown in **Figure 7**.

	Name	Taxonomy	Vancomycin_ARG
1	E1	s__Enterococcus_B hirae	0
2	E2	s__Enterococcus_B faecium	0
3	E3	s__Enterococcus_D casseliflavus	1
4	E4	s__Enterococcus_B hirae	1
5	E5	s__Enterococcus_B hirae	0
6	E6	s__Enterococcus faecalis	0
7	E7	s__Enterococcus faecalis	0
8	E8	s__Enterococcus faecalis	0
9	E9	s__Enterococcus faecalis	0
10	E10	s__Enterococcus faecalis	0
11	E11	s__Enterococcus faecalis	0
12	E12	s__Enterococcus faecalis	0
13	E13	s__Enterococcus faecalis	0
14	E14	s__Enterococcus faecalis	0
15	E15	s__Enterococcus faecalis	0
16	E16	s__Enterococcus faecalis	0
17	E17	s__Enterococcus faecalis	0
18	E18	s__Enterococcus faecalis	0
19	E19	s__Enterococcus faecalis	0
20	E20	s__Enterococcus faecalis	0
21	E21	s__Enterococcus faecalis	0
22	E22	s__Enterococcus faecalis	0
23	E23	s__Enterococcus faecalis	0
24	E24	s__Enterococcus faecalis	0
25	E25	s__Enterococcus faecalis	0
26	E26	s__Enterococcus_B lactis	1

Figure 7: One-Hot Encoded Table

Using the one-hot encoded data, a contingency table is constructed. After that, the association between Enterococcus's taxonomy and ARG is established using hypothesis testing, in particular the **Chi Square Test**. Following the indication of clear association, the second step would be to choose a machine learning-based classification model. In this research, **Naïve Bayes** is chosen to model the relationship as it offers clear interpretation for binary outcome variables (Brownlee, 2020).

4. Results

The contingency table, as shown in **Figure 8**, illustrates the taxonomic composition count as well as the ARG count in the samples. There are 20 samples, which are *Enterococcus faecalis*; 1 sample, which is *Enterococcus_B faecium*; 3 samples, which are *Enterococcus hirae*; 1 sample, which is *Enterococcus_B lactis*; and 1 sample, which is *Enterococcus_D casseliflavus*. On the other hand, there are 23 samples that don't carry the Vancomycin-resistant genes and 3 samples that do.

	Vancomycin_ARG Absent	Vancomycin_ARG Present
Enterococcus faecalis	20	0
Enterococcus_B faecium	1	0
Enterococcus_B hirae	2	1
Enterococcus_B lactis	0	1
Enterococcus_D casseliflavus	0	1

Figure 8: Contingency Table of Vancomycin ARG

Using the same contingency table, hypothesis testing (the Chi Square Test) shows that there is a strong association between Enterococcus's taxonomic composition and its ARG (Vancomycin resistance). This is evident by the p-value of 0.0006357, which is smaller than the significance value of 0.5. Hence, the null hypothesis of no association is rejected, resulting in the conclusion that there is a strong association.

The Naïve Bayes model is built using the one-hot encoded table, as shown in **Figure 7**. It is done by using the e1071 package in R Studio, as shown in **Figure 9**. The model input is Enterococcus's taxonomy, while the model output is the probability of the presence of Vancomycin-resistant genes.

```
# Naive Bayes Model
```{r}
library(e1071)
naive_bayes_model = naiveBayes(One_Hot_Encoded_Table$Vancomycin_ARG ~ One_Hot_Encoded_Table$Taxonomy, data = One_Hot_Encoded_Table)
```
```

Figure 9: Code Chunk for Naïve Bayes Model

To test the accuracy of the Naïve Bayes model, all five different Enterococcus's taxonomies are input as test data. Probability outputs were automatically converted to Boolean values based on the larger factor, as shown in **Figure 10**. Subsequently, the original dataset of 26 samples is also tested using this Naïve Bayes model, mimicking real-world scenarios where samples lack ARG indications. The resulting accuracy yielded 25 correct predictions and 1 incorrect prediction, with an overall accuracy rate of 96.15%.

Testing_Set <chr>	Class_Vancomycin_ARG_Absent <chr>	Class_Vancomycin_ARG_Present <chr>
Enterococcus faecalis	True	False
Enterococcus_B faecium	True	False
Enterococcus_B hirae	True	False
Enterococcus_B lactis	False	True
Enterococcus_D casseliflavus	False	True

5 rows

Figure 10: Test Data

5. Discussion

Considering the Naïve Bayes model's impressive accuracy of 96.15%, along with the strong association shown by a p-value of 0.0006357, there is promising potential for its application in real-world situations. However, several critical considerations in this model require attention:

1. **Limited Generalizability:** The testing sample exclusively comprises Enterococcus bacteria, which may restrict the model's applicability to other bacterial species. This limitation could potentially undermine its effectiveness in scenarios involving diverse microbial populations.
2. **Risk of Overfitting:** With only 26 samples, there is a risk of overfitting bias, where the model may have memorized noise from the training data rather than genuine patterns. This could compromise the model's ability to accurately generalize to new, unseen data.
3. **Bias Towards Vancomycin-resistant Genes:** The focus on Vancomycin-resistant genes in the ARG dataset may introduce bias and skew the model's predictions. By solely considering this subset of antibiotic resistance mechanisms, the model may overlook other critical resistance mechanisms, thereby limiting its clinical utility.

Hence, while the model demonstrates a good result and is suitable for this experiment and similar studies, these considerations must be acknowledged. Otherwise, the model might output a false positive or a false negative, which will significantly affect decision-making in the clinical context. To mitigate these biases, limitations, and assumptions, acquiring additional samples as training data is important. By expanding the training dataset, the model can better capture the complexities of microbial diversity and resistance profiles, thereby enhancing its reliability and applicability in real-world situations.

6. Conclusion

This research has demonstrated that the samples obtained from Lim Chu Kang open-caged sea farm mostly don't carry ARG (Vancomycin resistance). The contingency table, as shown in **Figure 8**, illustrates that out of 26 samples, 23 yield a negative result. Hence, this could be generalized to the whole farm, given that the sampling method used was random sampling.

Most importantly, the exploration into the **relationship between the taxonomy composition and the ARGs of Enterococcus sampled from Singapore's coastal water using machine learning** has provided valuable insights. The machine learning-based model is able to increase the efficiency of identifying AMR in bacteria while maintaining accuracy. However, as mentioned, it is also crucial to acknowledge that the model utilized in this study is subject to several considerations, limiting its viability for real-world applications.

Overall, this research highlights the importance of employing advanced computational methods to uncover the complexities of AMR dynamics in environmental samples. The insights gained from this study not only contribute to our understanding of AMR but also hold promise for informing future research directions, public health policies, and practical interventions aimed at mitigating the spread of antibiotic resistance. Moving forward, these findings could serve as a valuable foundation for the development of innovative technologies and strategies to combat AMR and safeguard public health.

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